Kirolos Morcos Hamdy

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SUMMARY

Computer and Communication Engineering student at Alexandria University with a strong interest in Backend Development and Software Engineering, supported by foundations in Cloud Computing and Computer Networking. Experienced with Python, Java, JavaScript, C, SQL, and database development through academic and personal projects. AWS Certified Cloud Practitioner with hands-on experience building software applications and embedded systems.

EDUCATION

Alexandria University – Faculty of Engineering

Bachelor of Computer and Communication Engineering, GPA: 3.6 / 4.0 Expected Graduation: 2027

PROJECTS

Isolated-Word Speech Recognition Using AVR ATmega32A

- Developed a complete hardware and software embedded speech recognition system using the AVR ATmega32A.
- Designed and built the hardware circuit and implemented embedded C firmware for real-time audio acquisition, preprocessing, MFCC feature extraction, and Dynamic Time Warping (DTW) classification.
- Integrated a microphone, external SPI SRAM, 16x2 LCD, LEDs, and UART, with Python scripts used to generate C-compatible speech feature templates.

Fast Typing Practice Website

- Developed an interactive typing practice website using HTML, CSS, and JavaScript.
- Implemented real-time typing speed (WPM) and accuracy calculations with an interactive user interface.
- Deployed the project using GitHub Pages.

Library Management System

- Developed the Backend of a Library Management System supporting admin and customer roles, managing books, publishers, orders, shopping carts, members, and transactions.
- Designed and implemented a relational SQL database with CRUD operations and RESTful APIs.
- Collaborated with a team to integrate the backend functionality with the overall application.

Connect Hub

- Contributed to a team-based Java social networking application inspired by social media platforms.
- Applied Object-Oriented Programming, Object-Oriented Design, SOLID principles, and design patterns to develop modular software components.
- Contributed primarily to User Account and Group Management components along with other system functionality.

Bank Management System

- Developed a console-based banking application in C for account management and common banking operations.
- Implemented file-based data storage, authentication, input validation, searching, sorting, deposits, withdrawals, transfers, and transaction reporting.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C, SQL
Backend & Web: REST APIs, Node.js, Express.js, Database Design, Object-Oriented Programming, HTML, CSS
Databases: MySQL
Cloud & Networking: AWS, CCNA-level Networking
Embedded Systems: AVR ATmega32A, ADC, SPI, UART, Embedded C
Tools: Git, GitHub, Linux, Cisco Packet Tracer, Postman, Wireshark, MATLAB

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02)** — Amazon Web Services (AWS)
- **Meta Back-End Developer Professional Certificate** — In Progress — 4 of 9 courses completed
- **CS50x: Introduction to Computer Science** — Harvard University
- **AWS 1: Intro to Cloud Computing** — Manara

TRAINING

- **CCNA Training** — WE Telecom Egypt